Отчёт по лабораторной работе №3

Агентное моделирование

Селиванов Вячеслав Алексеевич

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```
import Pkg
Pkg.activate("../project")
Pkg.instantiate()
```

```
  Activating project at `C:\Users\12232\Documents\GitHub\2026-
1--study--simulation-modeling\labs\lab03\project`
Precompiling packages...
   11161.8 ms  ✓ JumpProcesses
    9954.3 ms  ✓ StochasticDiffEq
   14926.9 ms  ✓ DifferentialEquations
  3 dependencies successfully precompiled in 40 seconds. 501 already pr
   1 dependency had output during precompilation:
┌ StochasticDiffEq
│  WARNING: Imported binding OrdinaryDiffEqCore.fixed_t_for_floatingp
└
```

1 Цель работы

Изучить и проанализировать агентный подход

2 Задание

Создать скрипты визуализирующие модель DaisyWorld

3 Теоретическое введение

Агентный подход к имитационному моделированию (Agent-Based Modeling, ABM) — это метод исследования сложных систем, в котором поведение системы возникает из взаимодействия множества автономных сущностей, называемых агентами. Вместо того чтобы описывать систему глобальными уравнениями, мы моделируем каждую индивидуальную единицу и правила её поведения, а затем наблюдаем, какие коллективные паттерны появляются снизу вверх. Этот подход особенно полезен, когда поведение системы трудно предсказать из-за нелинейностей, гетерогенности участников или адаптивных стратегий.

4 Выполнение лабораторной работы

Создадим рабочее пространство и инициализируем проект (рис. 4.1).

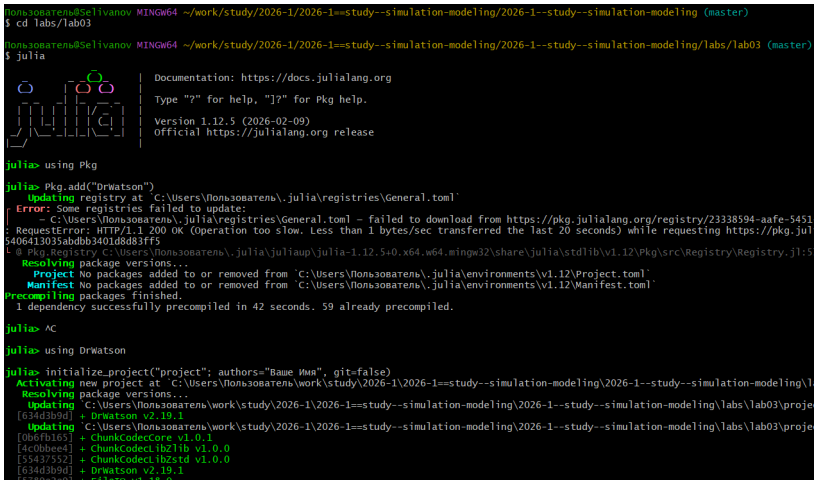

Рисунок 4.1: Инициализация проекта

Загрузим необходимые пакеты (рис. 4.2).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling
$ cd project

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling
$ julia add_packages.jl
   Installed OrdinaryDiffEqStabilizedIRK ──────── v1.9.0
   Installed OrdinaryDiffEqIMEXMultistep ──────── v1.10.0
   Installed OrdinaryDiffEqBDF ────────────────── v1.13.0
   Installed Quaternions ──────────────────────── v0.7.7
   Installed OrdinaryDiffEqHighOrderRK ────────── v1.8.0
   Installed OrdinaryDiffEqFeagin ─────────────── v1.7.0
   Installed StructArrays ─────────────────────── v0.7.2
   Installed OrdinaryDiffEqDefault ────────────── v1.11.0
   Installed OrdinaryDiffEqNordsieck ──────────── v1.7.0
    Updating `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-model
  [46ada45e] + Agents v7.0.0
  [6e4b80f9] + BenchmarkTools v1.6.3
  [336ed68f] + CSV v0.10.16
  [a93c6f00] + DataFrames v1.8.1
  [0c46a032] + DifferentialEquations v7.17.0
  [7073ff75] + IJulia v1.34.4
  [033835bb] + JLD2 v0.6.3
  [98b081ad] + Literate v2.21.0
  [91a5bcdd] + Plots v1.41.6
  [d7167be5] + Quarto v1.0.0
  [9a3f8284] + Random v1.11.0
    Updating `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-model
  [47edcb42] + ADTypes v1.21.0
  [1520ce14] + AbstractTrees v0.4.5
  [7d9f7c33] + Accessors v0.1.43
  [79e6a3ab] + Adapt v4.5.0
  [46ada45e] + Agents v7.0.0
  [66dad0bd] + AliasTables v1.1.3
  [a95523ee] + AlmostBlockDiagonals v0.1.10
  [ec485272] + ArnoldiMethod v0.4.0
  [4fba245c] + ArrayInterface v7.23.0
  [4c555306] + ArrayLayouts v1.12.2
```

Рисунок 4.2: Загрузка пакетов

В папке src создадим файл описывающий агента, далее в папке scripts создадим скрипт с предложенным кодом, выполняющим базовую визуализацию (рис. 4.3).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ julia --project=. scripts/daisyworld.jl
Шаг 1...
Шаг 6...
Шаг 46...
Готово! Графики сохранены в: C:\Users\Пользователь\work\study\2026-1\2026-1==stu
dy--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\pl
ots

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab03/project (master)
```

Рисунок 4.3: Выполненим базовую визуализацию

Создадим производные форматы (рис. 4.4).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ julia --project=. scripts/tangle.jl scripts/daisyworld.jl
Генерация из: scripts/daisyworld.jl
[ Info: generating plain script file from `C:\Users\Пользователь\work\study\2026
-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\la
b03\project\scripts\daisyworld.jl`
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study
--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scri
pts\daisyworld\daisyworld.jl`
 ✓ Чистый скрипт: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulat
ion-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisy
world/daisyworld.jl
[ Info: generating markdown page from `C:\Users\Пользователь\work\study\2026-1\2
026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\
project\scripts\daisyworld.jl`
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study
--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\mark
down\daisyworld\daisyworld.qmd`
 ✓ Quarto: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-mod
eling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld/
daisyworld.qmd
[ Info: generating notebook from `C:\Users\Пользователь\work\study\2026-1\2026-1
==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\proje
ct\scripts\daisyworld.jl`
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study
--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\note
books\daisyworld\daisyworld.ipynb`
 ✓ Notebook: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-m
odeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisywor
ld/daisyworld.ipynb

Готово! Все файлы созданы.
```

Рисунок 4.4: Создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.5).

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
include(srcdir("daisyworld.jl"))
using CairoMakie
model = daisyworld()
daisycolor(a::Daisy) = a.breed
plotkwargs = (
agent_color=daisycolor, agent_size = 20, agent_marker = '✿',
heatarray = :temperature,
heatkwargs = (colorrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)
save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)
[1]  ✓ 2m 25.5s

CairoMakie.Screen{IMAGE}
```

Рисунок 4.5: Проверка jupyter

Анимируем базовую визуализацию с помощью скрипта (рис. 4.6).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ julia --project=. scripts/daisyworld-animate.jl
Precompiling GeometryBasicsExt finished.
  1 dependency successfully precompiled in 10 seconds

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simula
$ |
```

Рисунок 4.6: Создание анимации

Выполним скрипт для визуализации динамики изменения числа маргариток(рис. 4.7).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ julia --project=. scripts/daisyworld-count.jl

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ |
```

Рисунок 4.7: Выполним визуализацию изменения количества маргариток

Создадим производные форматы (рис. 4.8).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/pro
$ julia --project=. scripts/tangle.jl scripts/daisyworld-count.jl
Генерация из: scripts/daisyworld-count.jl
[ Info: generating plain script file from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--si
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modelin
 ✓ Чистый скрипт: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\la
[ Info: generating markdown page from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simula
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modelin
 ✓ Quarto: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\pro
[ Info: generating notebook from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modelin
 ✓ Notebook: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\p

Готово! Все файлы созданы.
```

Рисунок 4.8: Создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.9).

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(srcdir("daisyworld.jl"))
using CairoMakie
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(; solar_luminosity = 1.0)
agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy count")
blackl = lines!(ax, agent_df[!, :time], agent_df[!, :count_black], color = :black)
whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white], color = :orange)
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsize = 12)
✓ 1m 5.6s

Legend()

figure

save(plotsdir("daisy_count.png"), figure)
✓ 4.3s

CairoMakie.Screen{IMAGE}
```

Рисунок 4.9: Проверка jupyter

Построим комплексный график числа маргариток, температуры и альбедо в зависимости от времени (рис. 4.10).

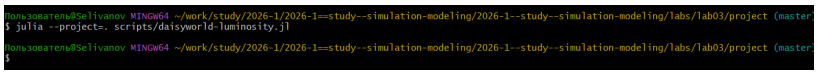

Рисунок 4.10: Построение комплексного графика

Создадим производные форматы (рис. 4.11).

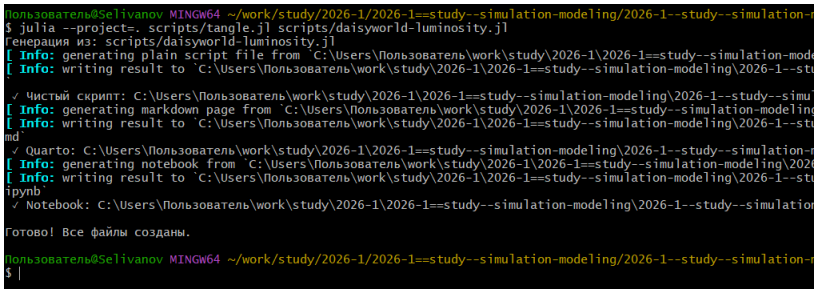

Рисунок 4.11: Создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.12).

```julia
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(srcdir("daisyworld.jl"))
using CairoMakie
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(solar_luminosity = 1.0, scenario = :ramp)
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]
agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)
figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
black1 = lines!(ax1, agent_df[!, :time], agent_df[!, :count_black], co
white1 = lines!(ax1, agent_df[!, :time], agent_df[!, :count_white], co
figure[1, 2] = Legend(figure, [black1, white1], ["black", "white"])
ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "luminosit
lines!(ax2, model_df[!, :time], model_df[!, :temperature], color = :re
lines!(ax3, model_df[!, :time], model_df[!, :solar_luminosity], color
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end
figure
save(plotsdir("daisy_luminosity.png"), figure)
```

[1] ✓ 1m 20.7s

```
CairoMakie.Screen{IMAGE}
```

Рисунок 4.12: Проверка jupyter

Выполним базовую визуализацию с параметрами (рис. 4.13).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/pro
$ julia --project=. scripts/daisyworld__param.jl

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/pro
$ |
```

Рисунок 4.13: Базовая визуализация с параметрами

Создадим производные форматы (рис. 4.14).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/projec
$ julia --project=. scripts/tangle.jl scripts/daisyworld__param.jl
Генерация из: scripts/daisyworld__param.jl
[ Info: generating plain script file from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simul
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\l
 ✓ Чистый скрипт: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03
[ Info: generating markdown page from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulatio
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\l
 ✓ Quarto: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\projec
[ Info: generating notebook from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-mode
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\l
 ✓ Notebook: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\proj

Готово! Все файлы созданы.

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/projec
$
```

Рисунок 4.14: Создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.15).

```julia
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
# Параметры эксперимента
param_dict = Dict(
:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)
# Создаём список всех комбинаций
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
daisycolor(a::Daisy) = a.breed
plotkwargs = (
agent_color=daisycolor,
agent_size = 20,
agent_marker = '✿',
heatarray = :temperature,
heatkwargs = (colorrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
```

✓ 1m 13.6s

Рисунок 4.15: Проверка jupyter

Выполним скрипт для визуализации динамики изменения числа маргариток с перебором параметров (рис. 4.16).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--si
$ julia --project=. scripts/daisyworld-count__param.jl

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--si
$
```

Рисунок 4.16: Визуализация изменения числа маргариток с перебором параметров

Создадим производные форматы (рис. 4.17).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-mode
$ julia --project=. scripts/tangle.jl scripts/daisyworld-count__param.jl
Генерация из: scripts/daisyworld-count__param.jl
[ Info: generating plain script file from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modelin
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study-
m.jl`
 ✓ Чистый скрипт: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulati
[ Info: generating markdown page from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\20
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study-
am.qmd`
 ✓ Quarto: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-mode
[ Info: generating notebook from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1-
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study-
ram.ipynb`
 ✓ Notebook: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-mo

Готово! Все файлы созданы.

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-mode
$
```

Рисунок 4.17: Создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.18).

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
# Параметры эксперимента
param_dict = Dict(
:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)
# Создаём список всех комбинаций
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabe
blackl = lines!(ax, agent_df[!, :time], agent_df[!, :co
✓ 1m 27.7s
```

Рисунок 4.18: Проверка jupyter

Построим комплексные графики числа маргариток, температуры и альбедо в зависимости от времени с перебором параметров, создадим производные форматы (рис. 4.19).

```
Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--
$ julia --project=. scripts/tangle.jl scripts/daisyworld-luminosity__param.jl
Генерация из: scripts/daisyworld-luminosity__param.jl
[ Info: generating plain script file from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--sim
l`
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling
sity__param.jl`
 ✓ Чистый скрипт: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--
am.jl
[ Info: generating markdown page from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulat
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling
osity__param.qmd`
 ✓ Quarto: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--
[ Info: generating notebook from `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-m
[ Info: writing result to `C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling
nosity__param.ipynb`
 ✓ Notebook: C:\Users\Пользователь\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study
ipynb

Готово! Все файлы созданы.

Пользователь@Selivanov MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--
$
```

Рисунок 4.19: Построение комплексных графиков с перебором параметров и создание производных форматов

Проверим работоспособность файла .ipynb (рис. 4.20).

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
# Параметры эксперимента
param_dict = Dict(
:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :ramp,
:seed => 165,
)
# Создаём список всех комбинаций
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
temperature(model) = StatsBase.mean(model.temperature
mdata = [temperature, :solar_luminosity]
agent_df, model_df = run!(model, 1000; adata = adata,
figure = CairoMakie.Figure(size = (600, 600));
1m 28.1s
```

Рисунок 4.20: Проверка jupyter

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
include(srcdir("daisyworld.jl"))
using CairoMakie
```

```
model = daisyworld()
daisycolor(a::Daisy) = a.breed
plotkwargs = (
agent_color=daisycolor, agent_size = 20, agent_marker = '□',
heatarray = :temperature,
heatkwargs = (colorrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs.
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs.
save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)
```

```
┌ Warning: DrWatson could not find find a project file by recursively cl
□ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

```
using DrWatson
@quickactivate "project"
using Agents
```

```
using DataFrames
using Plots
include(srcdir("daisyworld.jl"))
using CairoMakie
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(; solar_luminosity = 1.0)
agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy c
blackl = lines!(ax, agent_df[!, :time], agent_df[!, :count_black],
whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white],
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz
```

```
┌ Warning: DrWatson could not find find a project file by recursively c
⚠ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

```
Legend()
```

figure

```
save(plotsdir("daisy_count.png"), figure)
```

```
using DrWatson
@quickactivate "project"
using Agents
```

```julia
using DataFrames
using Plots
include(srcdir("daisyworld.jl"))
using CairoMakie
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
model = daisyworld(solar_luminosity = 1.0, scenario = :ramp)
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]
agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata
figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
blackl = lines!(ax1, agent_df[!, :time], agent_df[!, :count_black],
whitel = lines!(ax1, agent_df[!, :time], agent_df[!, :count_white],
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])
ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "lumino
lines!(ax2, model_df[!, :time], model_df[!, :temperature], color =
lines!(ax3, model_df[!, :time], model_df[!, :solar_luminosity], col
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end
figure
save(plotsdir("daisy_luminosity.png"), figure)
```

```
┌ Warning: DrWatson could not find find a project file by recursively c
⌧ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

```julia
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
# □□□□□□□□□ □□□□□□□□□□□□
param_dict = Dict(
:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)
# □□□□□□□ □□□□□□ □□□□ □□□□□□□□□□
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
daisycolor(a::Daisy) = a.breed
plotkwargs = (
```

```
agent_color=daisycolor,
agent_size = 20,
agent_marker = '⊠',
heatarray = :temperature,
heatkwargs = (colorrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs.
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs.
global counter
counter = counter + 1
plt1_name = "daisyworld_$(counter)_step01.png"
plt2_name = "daisyworld_$(counter)_step04.png"
plt3_name = "daisyworld_$(counter)_step40.png"
save(plotsdir(plt1_name), plt1)
save(plotsdir(plt2_name), plt2)
save(plotsdir(plt3_name), plt3)
end
```

```
┌ Warning: DrWatson could not find find a project file by recursively cl
⊠ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
```

```
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

```julia
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
black(a) = a.breed == :black
white(a) = a.breed == :white
```

```
adata = [(black, count), (white, count)]
# ⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧
param_dict = Dict(
:griddims => (30, 30),
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)
# ⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy c
blackl = lines!(ax, agent_df[!, :time], agent_df[!, :count_black],
whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white],
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz
global counter
counter += 1
```

```
plt_name = "daisy-count_$(counter).png"
save(plotsdir(plt_name), figure)
end
```

```
┌ Warning: DrWatson could not find find a project file by recursively c
⌧ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
using CairoMakie
include(srcdir("daisyworld.jl"))
black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]
# ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧
param_dict = Dict(
:griddims => (30, 30),
```

```
:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :ramp,
:seed => 165,
)
# □□□□□□□ □□□□□□ □□□□ □□□□□□□□□□
params_list = dict_list(param_dict)
counter = 0
for params in params_list
model = daisyworld(;params...)
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]
agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata
figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
blackl = lines!(ax1, agent_df[!, :time], agent_df[!, :count_black],
whitel = lines!(ax1, agent_df[!, :time], agent_df[!, :count_white],
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])
ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "lumino
lines!(ax2, model_df[!, :time], model_df[!, :temperature], color =
lines!(ax3, model_df[!, :time], model_df[!, :solar_luminosity], col
```

```julia
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end
global counter
counter +=1
plt_name = "daisy-luminosity_$(counter).png"
save(plotsdir(plt_name), figure)
end
```

```
┌ Warning: DrWatson could not find find a project file by recursively cl
⊠ (given dir: C:\Users\12232)
└ @ DrWatson C:\Users\12232\.julia\packages\DrWatson\2QF5p\src\proje
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
┌ Warning: Found `resolution` in the theme when creating a `Scene`. The
└ @ Makie C:\Users\12232\.julia\packages\Makie\kJl0u\src\scenes.jl:2
```

5 Выводы

В данной лабораторной работе я познакомился с агентным моделированием и моделью DaisyWorld, а так же построил их визуализацию.